



Avicenna's medical encyclopedia

Known as the "Prince of Physicians," the 11th-century Persian scholar Avicenna was a hugely influential figure both in the Middle East and in Europe, where his work *The Canon of Medicine* continued to be used as a standard medical text until the 16th century.

BEFORE

As the study of the Bible came to dominate intellectual activity in the West, knowledge of classical Greek learning declined.

CLASSICAL SCHOLARSHIP

Greek and Roman education included the teaching of **grammar, logic, and rhetoric** (the *trivium*) before moving on to **astronomy, arithmetic, music, and geometry** (the *quadrivium*). At Aristotle's Lyceum and Plato's Academy, students listened to their lecturers discuss philosophy and science.



ROMAN TEACHER AND PUPILS

DECAY OF CLASSICAL KNOWLEDGE

The classical approach continued into the early medieval period, but by now knowledge of the Greek language had largely died out in the West. **Christianity** was dominant, and education focused on **the Bible**. Some fragments of classical knowledge were preserved in texts such as the 6th century philosopher Boethius' *Consolation of Philosophy*. But by the 8th century, education in western Europe was structured almost entirely around the Bible, and scholarship had become largely cut off from Greek science and philosophy.

"Books have led some to **learning** and others to **madness**."

FRANCESCO PETRARCH, "REMEDIES FOR FORTUNE FAIR AND FOUL," 1365

Diffusion of Knowledge

Knowledge of classical Greek and Roman culture declined in Western Europe in the early medieval period, but survived in the East. Muslim scholars preserved the works of classical Greek philosophers and, as this knowledge spread west, laid the foundations for the European intellectual revival.

During the late 8th and 9th centuries, something of a literary awakening occurred at the court of the Frankish emperor Charlemagne (see pp.188–89). Before Charlemagne, society lacked many of the basic educational skills; most priests were barely literate and the royal court had difficulty finding educated men to act as scribes and copy out manuscripts. To fix these problems, Charlemagne created schools and assembled the greatest scholars of the age at his court. A Northumbrian scholar, Alcuin of York, was recruited as head of the palace school. A standard curriculum was introduced, and the classical study of the *trivium* and *quadrivium* was reestablished (see BEFORE). Latin once again became the formal language of communication across Europe.

The Arab inheritance

Charlemagne's reforms were primarily concerned with creating an educated clergy capable of reforming the Frankish church. Works of classical Greek science and philosophy remained essentially unavailable in the Christian Europe. In Arab Spain, North Africa, and the Middle East, however, the situation



Astrolabe manual

Muslim scholars are shown using the astrolabe, an instrument for locating and predicting the position of the Sun, Moon, and stars. The astrolabe was introduced into Christian Europe via Muslim Spain in the 11th century.

was very different. The Arab societies inhabiting these regions had inherited many aspects of classical Greek culture. Until the 12th century, for example, virtually all that Christian Europe knew of Aristotle came via Boethius (see BEFORE). To Islamic scholars, Aristotle was "the Philosopher" and his works were standard texts. In the 9th–10th centuries, numerous classical Greek works were translated into Arabic, and a flurry of commentaries were composed.

Two individuals stand out during this period. The Persian scholar Avicenna produced a huge range of material covering almost every area of knowledge, from metaphysics to medicine. His *Book of Healing* is the largest encyclopedia of knowledge composed by one person in the period. Averroes (see below), a renowned philosopher, scientist, and lawyer, composed a series of commentaries on Aristotle that for centuries formed the key source for Aristotle's philosophy in the West.

The 12th-century renaissance

The diffusion of knowledge from East to West accelerated during the 12th century. In Spain, the Christian reconquest of lands previously held by Muslim rulers encouraged the spread of Islamic learning. In the 13th century, the Spanish king Alfonso X established a program of translation of Greek and Arab texts in Toledo, previously capital of the Muslim caliphate of Cordoba.

Other important hubs for the exchange of ideas included the Christian crusader kingdoms that were established in the Middle East in the 12th–13th centuries (see pp.200–201) and the Greek empire of Byzantium (see pp.198–99), which fell into the hands of the crusaders in 1204. By the 13th century, Latin translations of the most important classical texts were available to European scholars.

ARAB PHILOSOPHER AND SCIENTIST (1126–98)

AVERROES

Born into a learned and cultured family in the city of Cordoba (then part of Muslim Spain) Averroes was an expert in Islamic law and theology, as well as Arabic grammar and poetry. In 1153, he was invited to the court of the Almohad caliphs in Morocco, where he worked as a judge and physician, and wrote important works on medicine, philosophy, and law. He later fell out of favor and was exiled. Averroes' writings had a huge influence on European scholarship, but his greatest contribution was a series of commentaries on Aristotle that introduced medieval scholars to the works of the great Greek philosopher.



Scholasticism

The rediscovery of ancient philosophical works combined with the ongoing development of Christian theology led to the scholastic movement of the later Medieval period. Scholasticism was not a philosophy in itself, but rather a tool for learning that placed emphasis on logic and reasoning. The works of Plato

RHETORIC Derived from the Greek word for "orator," rhetoric is the art of persuasion through spoken or written language. It was an important art in classical Greece, where public speaking was central to political life.